

tf.compat.v1.metrics.precision_at_k Stay organized with collections Save and categorize content based on your preferences.

https://www.tensorflow.org/api_docs/python/tf/compat/v1/metrics/precision_at_k

Computes precision@k of the predictions with respect to sparse labels.

Function Signatures

```
tf.compat.v1.metrics.precision_at_k( labels, predictions, k,  
class_id=None, weights=None, metrics_collections=None,  
updates_collections=None, name=None )
```

Code Examples

```
tf.compat.v1.metrics.precision_at_k(  
    labels,  
    predictions,  
    k,  
    class_id=None,  
    weights=None,  
    metrics_collections=None,  
    updates_collections=None,  
    name=None  
)
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    labels,  
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)
```

Documentation

Computes precision@k of the predictions with respect to sparse labels.

If `class_id` is specified, we calculate precision by considering only the entries in the batch for which `class_id` is in the top-k highest predictions, and computing the fraction of them for which `class_id` is indeed a correct label.

If `class_id` is not specified, we'll calculate precision as how often on average a class among the top-k classes with the highest predicted values

of a batch entry is correct and can be found in the label for that entry.

`precision_at_k` creates two local variables, `true_positive_at_` and `false_positive_at_`, that are used to compute the `precision@k` frequency. This frequency is ultimately returned as `precision_at_`: an idempotent operation that simply divides `true_positive_at_` by total (`true_positive_at_ + false_positive_at_`).

For estimation of the metric over a stream of data, the function creates an `update_op` operation that updates these variables and returns the `precision_at_`. Internally, a `top_k` operation computes a Tensor indicating the top k predictions. Set operations applied to `top_k` and labels calculate the true positives and false positives weighted by weights. Then `update_op` increments `true_positive_at_` and `false_positive_at_` using these values.

If `weights` is `None`, `weights` default to 1. Use weights of 0 to mask values.

Returns

Raises